جــامعـــة الأقـصــى

Al-Aqsa University
Faculty of Information Technology

2025-2026

First Semester

Information Retrieval

Project:

Text Search Engine using TF-IDF

Instructor:

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The submission deadline is on:

31-01-2026 11:59 PM

Text Search Engine using TF-IDF

Project Description

In this project, students will build a simple text-based information retrieval system using TF-IDF and Cosine Similarity. The goal is to understand how textual data can be converted into numerical representations and used to rank documents based on relevance to a user query.

Project Objectives

- Understand the concept of TF-IDF in Information Retrieval
- Apply text preprocessing techniques
- Build a basic search engine using Python
- Analyze and interpret retrieval results

Dataset Requirements

Use a dataset containing 20 to 30 text documents. The documents can be short articles,

educational paragraphs, or any text-based content. If an external dataset is used, the source must be clearly mentioned.

Text Preprocessing

- Convert text to lowercase
- Remove punctuation
- Remove English stop words

TF-IDF Model

Use TfidfVectorizer from scikit-learn to convert documents into TF-IDF vectors.

Search and Ranking

The system should accept a user query, transform it using the same TF-IDF model, compute

Cosine Similarity between the query and all documents, and return the top five most relevant

documents.

Analysis and Discussion

- Why did the top-ranked document achieve the highest similarity score?
- Which terms contributed most to the similarity score?
- What is the effect of removing stop words?

Deliverables

- A 2 versions of Google Colab notebook (Pdf and ipynb) that containing:
    - First cell with your name and ID
    - The full implementation.
    - Well-documented Python code with comments.
- A short written report (1–2 pages)
- Place the three files in a folder with the student's name and submit it as a ZIP file.

Evaluation Criteria

- Correct implementation of TF-IDF (30%)
- Search and ranking functionality (25%)
- Analysis and interpretation of results (25%)
- Code organization and clarity (20%)

Important Notice:

Any project that is found to be entirely copied will result in the grade being **equally divided among all students involved**, regardless of the number of students. Academic integrity is essential, and originality of work is strictly required.